

Artificial Intelligence Basics Course Summary

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1 Key Terms

1.1 Search Algorithms

- **Minimax Search:** A search algorithm for two-player zero-sum games that aims to find the best move for the current player, assuming the opponent also plays optimally.
- **Alpha-Beta Pruning:** A technique to reduce the search space in Minimax search by avoiding the evaluation of branches that cannot affect the optimal decision.
- **Game Tree:** A tree-like structure representing all possible sequences of moves in a game, with leaf nodes representing the outcomes.

1.2 Linear Algebra

- **Matrix:** A rectangular array of numbers, symbols, or expressions, arranged in rows and columns.
- **Linear Transform:** A function that maps vectors from one vector space to another, satisfying $T(u+v) = T(u) + T(v)$ and $T(cu) = cT(u)$ for all vectors u, v and scalars c .
- **Matrix Multiplication:** An operation that combines two matrices to produce a third matrix, where the element at row i and column j of the resulting matrix is the dot product of the i -th row of the first matrix and the j -th column of the second matrix. The inner dimensions must match.
- **Matrix Inverse:** For a square matrix A , its inverse A^{-1} is a matrix such that $AA^{-1} = A^{-1}A = I$, where I is the identity matrix.
- **Matrix Transposition:** An operation that swaps the rows and columns of a matrix. The transpose of matrix A is denoted as A^T .
- **Derivative:** A measure of how a function changes with respect to a change in its input.
- **Partial Derivative:** A derivative of a function of multiple variables with respect to one of those variables, holding the others constant.
- **Sigmoid Function:** A function that maps any real number to a value between 0 and 1, often used as an activation function in neural networks. Its formula is $\sigma(x) = \frac{1}{1+e^{-x}}$.

1.3 Neural Networks

- **Perceptron:** A fundamental unit in neural networks that takes multiple inputs, applies weights to them, sums the weighted inputs, and produces an output based on a threshold.

1.4 Artificial Intelligence Fundamentals

- **Turing Test:** An operational test for intelligent behavior, the Imitation Game, where a human interrogator tries to distinguish between a human and a machine based on their text-based responses.
- **Inductive Reasoning:** The process of deriving general principles from specific observations.
- **Linear Regression:** A machine learning algorithm used to model the relationship between a dependent variable and one or more independent variables by fitting a linear equation to observed data.
- **Model:** A mathematical representation of a system or process, used to make predictions or understand relationships between variables.
- **Overfitting:** A phenomenon in machine learning where a model learns the training data too well, resulting in poor generalization to unseen data.
- **Loss Function:** A function that quantifies the difference between the predicted and actual values in a machine learning model.
- **Decision Boundary:** A line or surface that separates different classes of data points in a classification problem.
- **Stochastic Gradient Descent:** An iterative optimization algorithm that updates model parameters using the gradient calculated from a randomly selected subset of the training data.
- **Logistic Regression:** A statistical model that uses a logistic function to model the probability of a binary outcome.
- **Empirical Risk Minimizer:** A learning principle that aims to find the model that minimizes the average loss over the training data.
- **Perceptron Algorithm:** An iterative algorithm for binary classification that updates the weights of a linear model based on misclassified data points.

1.5 Reinforcement Learning

- **Reinforcement Learning:** A type of machine learning where an agent learns to make decisions by interacting with an environment and receiving rewards or penalties.

2 Problem Types

2.1 Foundational AI Concepts

- **Understanding the Nature of Intelligence:** This problem explores the definition and characteristics of intelligence, both human and artificial. It examines various perspectives on what constitutes intelligence and how it can be measured or simulated.
- **Defining Artificial Intelligence:** This problem involves formulating a precise definition of artificial intelligence, considering different perspectives and historical contexts. The challenge lies in capturing the essence of AI while accounting for its evolving nature and diverse applications.
- **Passing the Turing Test:** This problem focuses on the implications of passing the Turing Test, questioning whether passing the test truly indicates machine intelligence or merely the ability to mimic human behavior. It also considers counterarguments, such as Searle's Chinese Room argument.
- **Overcoming the Limitations of Search:** This problem addresses the computational limitations of exhaustive search in complex games like chess and Go. It highlights the need for alternative approaches, such as machine learning, to handle the vast search spaces involved.

- **Understanding the Differences Between Game Types:** This problem involves comparing and contrasting different game types, such as chess and soccer, to understand the unique challenges they present for AI algorithms. It explores how the complexity and dynamics of different games influence the design of AI systems.
- **Exploring the Similarities Between AI Tasks:** This problem involves identifying commonalities between seemingly disparate AI tasks, such as speech synthesis and game playing. It explores the underlying principles and techniques that can be applied across different AI domains.

2.2 Machine Learning Algorithms

- **Addressing the Challenges of Computer Vision:** This problem explores the difficulties in developing computer vision systems that can reliably interpret and understand images. It contrasts the initial optimism about the tractability of computer vision with the persistent challenges that remain decades later.
- **Developing Effective Game-Playing Algorithms:** This problem focuses on the design and implementation of algorithms for playing games. It explores the key components of such algorithms, including state representation, action selection, and opponent modeling.
- **Understanding Perceptrons:** This problem involves understanding the fundamental building blocks of neural networks, namely perceptrons, and how they process information.
- **Understanding Reinforcement Learning:** This problem involves understanding the principles of reinforcement learning and how it can be applied to game playing, using the example of Samuel's checkers program.
- **Understanding the AlphaGo System:** This problem involves understanding the architecture and functionality of the AlphaGo system, which combines deep learning and search techniques to play the game of Go.
- **Linear Regression:** Finding the line that best fits a set of data points, minimizing the sum of squared distances between the points and the line.
- **Multiple Linear Regression:** Extending linear regression to include multiple independent variables to predict a dependent variable.
- **Curve Fitting:** Fitting a curve to a set of data points, potentially using polynomial functions of higher degrees.
- **Binary Classification:** Classifying data points into two categories (+1 or -1) using a linear model and a suitable loss function.
- **Overfitting:** The phenomenon where a model fits the training data too closely, resulting in poor generalization to unseen data.
- **Choosing a Loss Function:** Selecting an appropriate loss function (e.g., 0-1 loss, perceptron loss, hinge loss, logistic loss) for binary classification, considering differentiability and computational efficiency.
- **Minimizing Loss using Gradient Descent:** Employing gradient descent or stochastic gradient descent to iteratively update model parameters and minimize the chosen loss function.

2.3 AI Applications \& Systems

- **Understanding the AI Stack:** This problem involves understanding the different layers and components that constitute the AI stack, including machine learning, search algorithms, computer vision, and human-AI interaction.

2.4 Linear Algebra for AI

- **Representing Linear Transformations with Matrices:** Matrices are used to represent linear transformations, which map vectors from one space to another while preserving linear combinations. This is fundamental to many AI algorithms.
- **Matrix Arithmetic:** The slide covers basic matrix operations such as addition, subtraction, scalar multiplication, and multiplication, which are essential for manipulating data in AI applications.
- **Solving Systems of Linear Equations:** Systems of linear equations can be represented and solved using matrices and their inverses. This is crucial for various AI tasks involving optimization and data analysis.
- **Matrix Transposition:** Matrix transposition is a fundamental operation that involves swapping rows and columns of a matrix. This is used for data manipulation and transformation in AI.
- **Parallelizing Matrix Multiplication:** The slide demonstrates how matrix multiplication can be parallelized using domain decomposition, improving computational efficiency in AI applications.

2.5 Calculus for AI

- **Calculus in AI:** The slide briefly introduces the importance of calculus, specifically derivatives, partial derivatives, and finding maxima/minima/gradients, in AI algorithms.
- **Calculating the Derivative of the Sigmoid Function:** The slide shows the step-by-step derivation of the derivative of the sigmoid function using calculus rules. This is important for training neural networks.

2.6 AI Safety and Ethics

- **Mitigating AI Safety Risks:** This problem focuses on the potential risks associated with AI systems, such as vulnerabilities to adversarial attacks and the presence of biases in algorithms. It explores methods for improving AI safety and mitigating these risks.
- **Addressing Ethical Concerns in AI:** This problem examines the ethical implications of AI systems, particularly concerning bias and fairness. It highlights the importance of addressing these issues to ensure responsible AI development and deployment.

3 Algorithm Solutions

3.1 Fundamental Linear Algebra

- **Matrix Addition:** $A + B = C$, where $c_{ij} = a_{ij} + b_{ij}$
- **Matrix Scalar Multiplication:** $k \cdot A = B$, where $b_{ij} = k \cdot a_{ij}$
- **Matrix Multiplication:** $C_{ij} = \sum_{k=1}^n a_{ik} b_{kj}$
- **Matrix Transposition:** Swapping rows and columns of a matrix A to obtain A^T .
- **Solving Linear Equations with Invertible Matrices:** $Ax = b \Rightarrow x = A^{-1}b$

3.2 Calculus for Machine Learning

- **Derivative of Sigmoid Function:** $\frac{d}{dx} \sigma(x) = \frac{e^{-x}}{(1+e^{-x})^2}$

3.3 Supervised Machine Learning Algorithms

3.4 Artificial Neural Networks (ANN) Fundamentals

- **Perceptron Algorithm:** Iteratively update the weight vector w based on misclassified data points. If $\text{sign}(w^T x_n) \neq y_n$, then $w \leftarrow w + y_n x_n$.

3.5 Game Playing Algorithms

- **Minimax Search:** A recursive algorithm for two-player zero-sum games that explores the game tree to find the best move for the current player, assuming the opponent also plays optimally. It alternates between maximizing the score for the current player and minimizing the score for the opponent.
- **Alpha-Beta Pruning:** An optimization technique for Minimax search that reduces the search space by avoiding the evaluation of branches that cannot affect the optimal decision. It uses alpha and beta values to prune branches.